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End-Semester Examination (Total Marks 50)
PH 102 - PHYSICS II
2-5 PM, May 3, 2016

Read these instructions carefully

- a) All questions are compulsory.
- b) You **must** write all parts of the answers in the space provided for the given question. It is *advised* that you solve the problems in the roughbook (usual answerbook) and then copy the key steps in the space provided for that problem.
- c) Write **main key steps** instead of just answers. There is partial credit for intermediate steps. Just answers without justifications may not be awarded marks.
- d) **Write your name and roll number on each page.**

Useful Formulae:

a) $\nabla \cdot \mathbf{A} = \frac{1}{r^2} \frac{\partial (r^2 A_r)}{\partial r} + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (A_\theta \sin \theta) + \frac{1}{r \sin \theta} \frac{\partial A_\varphi}{\partial \varphi}$ in spherical coordinates.

b) $\nabla \cdot \mathbf{A} = \frac{1}{\rho} \frac{\partial (\rho A_\rho)}{\partial \rho} + \frac{1}{\rho} \frac{\partial A_\varphi}{\partial \varphi} + \frac{\partial A_z}{\partial z}$ in cylindrical coordinates.

c) $\nabla \times \mathbf{A} = \frac{1}{\rho} \begin{bmatrix} \hat{\rho} & \rho \hat{\phi} & \hat{z} \\ \frac{\partial}{\partial \rho} & \frac{\partial}{\partial \phi} & \frac{\partial}{\partial z} \\ A_\rho & \rho A_\phi & A_z \end{bmatrix}$ in cylindrical coordinates.

- d) Electric and magnetic fields produced by electric dipole $\mathbf{p}$ and magnetic dipole $\mathbf{m}$ are

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0 r^3} [3(\mathbf{p} \cdot \hat{r}) \hat{r} - \mathbf{p}] \text{ and } \mathbf{B} = \frac{\mu_0}{4\pi r^3} [3(\mathbf{m} \cdot \hat{r}) \hat{r} - \mathbf{m}], \text{ respectively.}$$

- e) The solution to the second order inhomogeneous linear differential equation of type

$$\frac{d^2 x(t)}{dt^2} + \omega^2 x(t) = B$$

is $x(t) = B/\omega^2 + C_1 \sin(\omega t) + C_2 \cos(\omega t)$ where B, C_1, C_2 and ω are constants.

Examiner's Signature

Student's Signature

Invigilator's Signature

Question 1: Answer the following short questions:

- a) Use divergence theorem for the vector field $\mathbf{v} = \mathbf{c}T$, where $\mathbf{c}$ is a constant vector and T is a scalar field, to prove

$$\int_V (\nabla T) d\tau = \oint_S T d\mathbf{a}.$$

The surface S encloses the volume V and $d\mathbf{a}$ is elemental vector area. [2]

Solution:

By divergence theorem

$$\begin{aligned} \int_V (\nabla \cdot (\mathbf{c}T)) d\tau &= \oint_S \mathbf{c}T \cdot d\mathbf{a} \\ \int_V (\mathbf{c} \cdot \nabla T + T \nabla \cdot \mathbf{c}) d\tau &= \mathbf{c} \cdot \oint_S T d\mathbf{a} \\ \mathbf{c} \cdot \int_V \nabla T d\tau &= \mathbf{c} \cdot \oint_S T d\mathbf{a} \quad \because \nabla \cdot \mathbf{c} = 0 \end{aligned}$$

Since $\mathbf{c}$ is completely arbitrary vector (though constant) field, the required identity is true. [Alternatively, choose $\mathbf{c} = \hat{\mathbf{i}}$ to prove x component of the identity and similarly the other components]

- b) Find the charge and current distribution represented by the scalar and vector potentials given by $\Phi(\mathbf{r}, t) = 0$, $\mathbf{A}(\mathbf{r}, t) = -\frac{qt}{4\pi\epsilon_0 r^2} \hat{\mathbf{r}}$. [2]

Solution:

Electric Field $\mathbf{E} = -\nabla\Phi - \frac{\partial\mathbf{A}}{\partial t} = \frac{q}{4\pi\epsilon_0} \frac{\hat{\mathbf{r}}}{r^2}$, magnetic field induction $\mathbf{B} = \nabla \times \mathbf{A} = 0$. These are the fields produced by a stationary point charge q at the origin.

- c) The electric potential at a point, is given by

$$V(r, \theta) = \begin{cases} \frac{k}{3\epsilon_0} r \cos \theta & r < R \\ \frac{k}{3\epsilon_0} \frac{R^3}{r^2} \cos \theta & r > R \end{cases}$$

where r and θ are spherical coordinates of the point. k and R are positive real constants. Find surface charge density at the surface $r = R$. [2]

Solution:

The surface charge density is given by

$$\begin{aligned} \sigma(\theta) &= -\epsilon_0 \left. \frac{\partial}{\partial r} V(r > R) \right|_{r=R} + \epsilon_0 \left. \frac{\partial}{\partial r} V(r < R) \right|_{r=R} \\ &= -\frac{k}{3} \cos \theta \left(-\frac{2R^3}{r^3} \right)_{r=R} + \frac{k}{3} \cos \theta \\ &= k \cos \theta \end{aligned}$$

- d) A spherical conductor of radius a , carries a charge Q . It is surrounded by a linear dielectric material of susceptibility χ_e , with outer radius b . Find the electrostatic energy of this configuration. [2]

Solution:

The Electric field $E(r) = \begin{cases} \frac{Q}{4\pi\epsilon_0 r^2} \hat{\mathbf{r}} & b < r \\ \frac{Q}{4\pi\epsilon r^2} \hat{\mathbf{r}} & a < r < b \end{cases}$. The energy of this system $W = \frac{1}{2} \int D \cdot E dv = \frac{Q}{8\pi\epsilon_0} \left[\frac{1}{(1+\chi_e)} \left(\frac{1}{a} - \frac{1}{b} \right) + \frac{1}{b} \right]$

- e) Consider a point charge q placed at the origin of an infinite dielectric medium of permittivity ($\epsilon = \epsilon_0 \epsilon_r$). Find the polarization $\mathbf{P}$, the polarization (bound) charges and the total charge. [2]

Solution:

Gauss Law: $\nabla \cdot \mathbf{D} = \rho_f \Rightarrow \mathbf{D} = \frac{q}{4\pi r^2} \hat{r}$, polarization $\mathbf{P} = \mathbf{D} - \epsilon_0 \mathbf{E} \Rightarrow \mathbf{P} = \frac{q}{4\pi r^2} \left(1 - \frac{\epsilon_0}{\epsilon}\right) \hat{r}$
Volume bound charge $\rho_b = -\nabla \cdot \mathbf{P} = -q \left(1 - \frac{\epsilon_0}{\epsilon}\right) \delta^3(\mathbf{r})$, Total charge $Q = -q \left(1 - \frac{\epsilon_0}{\epsilon}\right) + q = \frac{q}{\epsilon_r}$

- f) A region $z > 0$ has $\mu_r = 4$, while the region $z < 0$ has $\mu_r = 1$. The magnetic field $\mathbf{B}$ is uniform for $z > 0$ and is given by $\mathbf{B} = \frac{B_0}{2} \left(\sqrt{\frac{3}{2}} \hat{x} + \sqrt{\frac{3}{2}} \hat{y} + \hat{z} \right)$. Find $\mathbf{B}$ and $\mathbf{H}$ for $z < 0$ assuming that there is no free current at the interface $z = 0$. [2]

Solution:

By using boundary conditions, we get

$$B_z(z < 0) = \frac{B_0}{2}$$

and

$$B_{\parallel}(z < 0) = \frac{B_0}{8} \left(\sqrt{\frac{3}{2}} \hat{x} + \sqrt{\frac{3}{2}} \hat{y} \right)$$

Thus

$$\mathbf{B}(z < 0) = \frac{B_0}{8} \left(\sqrt{\frac{3}{2}} \hat{x} + \sqrt{\frac{3}{2}} \hat{y} + 4\hat{z} \right)$$

and

$$\mathbf{H} = \mathbf{B}/\mu_0.$$

- g) In *free space*, the magnetic field is given by

$$\mathbf{H} = r \left(\sin \phi \hat{r} + 2 \cos \phi \hat{\phi} \right) \cos(\omega t)$$

where r and ϕ are cylindrical coordinates and ω is a constant. Find the corresponding displacement current density $\mathbf{J}_D$. [2]

Solution:

magnetic field $\mathbf{H} = r \left(\sin \phi \hat{r} + 2 \cos \phi \hat{\phi} \right) \cos(\omega t)$, displacement current density $\mathbf{J}_D = \nabla \times \mathbf{H} = 3\hat{z} \cos \phi \cos \omega t$

- h) In a dielectric medium ($z < 0$, $\epsilon = 9\epsilon_0$, $\mu = \mu_0$) a plane wave with

$$\mathbf{E} = 8 \left(\sqrt{8} \hat{x} - \hat{z} \right) e^{-i(10^9 t - bx - \sqrt{8}bz)}$$

is incident on the boundary with air at $z = 0$. Find the angle of reflection θ_R and the angle of transmission θ_T . b is a positive constant. [2]

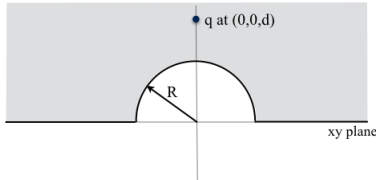
Solution:

Electric field $E = 8 \left(\sqrt{8} \hat{x} - \hat{z} \right) e^{-i(10^9 t - \zeta x - \sqrt{8}\zeta z)}$ $\tan \theta_I = \frac{k_x^I}{k_z^I} = \frac{b}{\sqrt{8}b} \Rightarrow \theta_I = \tan^{-1} \left(\frac{1}{\sqrt{8}} \right) = 19.47$ and from Snell's law $\sin \theta_T = \frac{n_1}{n_2} \sin \theta_I = 1 \Rightarrow \theta_T = \pi/2$.

Question 2: A conducting grounded sheet lies in the xy plane with a hemispherical boss (radius R) centered at the origin (see figure). A point charge q is placed at the distance $d(> R)$ on the z axis.

[2+1+3]

- Write down the image charges and their locations. (Don't derive, just guess on the basis of the classroom examples and write down the answer.)
- Write down the potential at a general point $\mathbf{r} \equiv (x, y, z)$ in the shaded region ($z > 0$ and $r > R$).
- Find the induced surface charge density and total charge induced on the *flat surface* of the conductor.



Solution:

- $q_1 = -q$ at $(0, 0, d)$, $q_{2/3} = \mp q \frac{R}{d}$ at $(0, 0, \pm \frac{R^2}{d})$ 2 marks here.

- The potential is

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \left[\frac{q}{(x^2 + y^2 + (z-d)^2)^{1/2}} + \frac{(-q)}{(x^2 + y^2 + (z+d)^2)^{1/2}} + \frac{(-qR/d)}{(x^2 + y^2 + (z - R^2/d)^2)^{1/2}} + \frac{(qR/d)(R^2/d)}{(x^2 + y^2 + (z + R^2/d)^2)^{1/2}} \right]$$

- Take a point on the flat section of the conductor. The coordinates are $(x, y, 0)$. The induced charge density is

$$\begin{aligned} \sigma(x, y) &= \epsilon_0 \mathbf{E} \cdot \hat{\mathbf{z}} \\ &= \frac{1}{4\pi} \left[\frac{q(-d)}{(x^2 + y^2 + d^2)^{3/2}} + \frac{(-q)d}{(x^2 + y^2 + d^2)^{3/2}} + \frac{(-qR/d)(-R^2/d)}{(x^2 + y^2 + R^4/d^2)^{3/2}} + \frac{(qR/d)(R^2/d)}{(x^2 + y^2 + R^4/d^2)^{3/2}} \right] \\ &= \frac{q}{4\pi} \left[-\frac{2d}{(x^2 + y^2 + d^2)^{3/2}} + \frac{2R^3}{d^2 (x^2 + y^2 + R^4/d^2)^{3/2}} \right] \end{aligned}$$

And the total charge

$$\begin{aligned} Q_{ind} &= \frac{q}{4\pi} \left[-\int_R^\infty \frac{2d(2\pi s ds)}{(s^2 + d^2)^{3/2}} + \int_R^\infty \frac{2R^3(2\pi s ds)}{d^2 (x^2 + y^2 + R^4/d^2)^{3/2}} \right] \\ &= q \left[-d \frac{1}{\sqrt{R^2 + d^2}} + \frac{R^3}{d^2 \sqrt{R^2 + R^4/d^2}} \right] \\ &= -\frac{q(d^2 - R^2)}{d\sqrt{R^2 + d^2}} \end{aligned}$$

Question 3: An electron is injected with a velocity $\mathbf{u}_0 = \hat{y} u_0$ at the origin into a region where an electric field $\mathbf{E} = \hat{z} E_0$ and a magnetic field $\mathbf{B} = \hat{x} B_0$, both exist. [3+3+1]

a) Write down the equation of motion of the electron.

b) Obtain the trajectory equation.

c) What is the trajectory if $E_0 = u_0 B_0$?

Solution:

a) By Lorentz force law (electronic charge, $-e$)

$$\begin{aligned} m\dot{\mathbf{v}} &= -e(\hat{\mathbf{z}}E_0 + \mathbf{v} \times \hat{\mathbf{x}}B_0) \\ &= -(eE_0 - ev_y B_0)\hat{\mathbf{z}} - ev_z B_0 \hat{\mathbf{y}} \end{aligned}$$

Then, clearly $v_x = 0$ at all times and

$$\begin{aligned} \dot{v}_y &= -\frac{eB_0}{m}v_z \\ \dot{v}_z &= -\frac{eE_0}{m} + \frac{eB_0}{m}v_y \end{aligned}$$

b) Taking one more derivative of v_z , we get (let $\omega = eB_0/m$)

$$\ddot{v}_z = \omega \dot{v}_y = -\omega^2 v_z.$$

Using initial conditions $v_z = 0$ and $\dot{v}_z = -\frac{eE_0}{m} + \frac{eB_0}{m}u_0$ at $t = 0$,

$$v_z = A \sin \omega t$$

where

$$A = -\frac{E_0}{B_0} + u_0.$$

Putting in the first equation and using initial condition,

$$\begin{aligned} \dot{v}_y &= -A\omega \sin \omega t \\ \Rightarrow v_y &= \left(u_0 - \frac{E_0}{B_0}\right) \cos \omega t + \frac{E_0}{B_0}. \end{aligned}$$

Then,

$$\begin{aligned} x(t) &= 0, \\ y(t) &= \frac{E_0}{B_0}t + \frac{1}{\omega} \left(u_0 - \frac{E_0}{B_0}\right) \sin \omega t, \\ z(t) &= \frac{1}{\omega} \left(u_0 - \frac{E_0}{B_0}\right) (1 - \cos \omega t). \end{aligned}$$

Thus, the trajectory is

$$\left(y - \frac{E_0}{B_0}t\right)^2 + \left(z - \frac{1}{\omega} \left(u_0 - \frac{E_0}{B_0}\right)\right)^2 = \frac{1}{\omega^2} \left(u_0 - \frac{E_0}{B_0}\right)^2.$$

c) If $u_0 = E_0/B_0$, then the trajectory is $y(t) = E_0 t/B_0$.

Question 4: A sphere of radius R has the magnetization $\mathbf{M} = \hat{\mathbf{z}} a (\hat{\mathbf{z}} \cdot \mathbf{r})^2$ where a is a constant. Find the following: [1+1+3+2]

- a) Bound volume current density in the sphere.
- b) Bound surface current density on the sphere.
- c) Magnetic moment of the sphere.
- d) Magnetic vector potential $\mathbf{A}$ at a distance $r \gg R$.

Solution:

- a) Note that $\mathbf{M} = az^2\hat{\mathbf{z}}$. Then, the bound volume current density $\mathbf{J}_b = \nabla \times \mathbf{M} = 0$.
- b) The bound surface current density $\mathbf{K}_b = \mathbf{M} \times \hat{\mathbf{n}}$. Here $\hat{\mathbf{n}} = \hat{\mathbf{r}}$. Then $\mathbf{K}_b = aR^2 \cos^2 \theta \sin \theta \hat{\phi}$.
- c) Consider a ring of thickness $Rd\theta$ in a plane perpendicular to z axis. The current through the ring is $dI = K_b R d\theta = aR^3 \cos^2 \theta \sin \theta$. The magnetic moment of this ring is

$$d\mathbf{m} = \pi (R \sin \theta)^2 dI \hat{\mathbf{z}}$$

And the total moment of the sphere would be

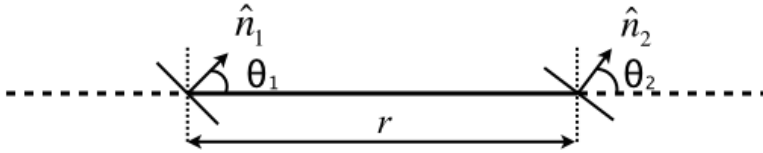
$$\begin{aligned} \mathbf{m} &= \pi a R^5 \int_0^\pi \cos^2 \theta \sin^3 \theta d\theta \hat{\mathbf{z}} \\ &= \frac{4}{15} \pi a R^5 \hat{\mathbf{z}} \end{aligned}$$

- d) Magnetic vector potential at a distance $r \gg R$ is given by

$$\begin{aligned} \mathbf{A} &= \frac{\mu_0}{4\pi} \frac{\mathbf{m} \times \mathbf{r}}{r^3} \\ &= \frac{\mu_0}{15} \frac{aR^5}{r^2} \sin \theta \hat{\phi} \end{aligned}$$

Question 5: Two plane currents loops, each with area A and carrying current I , are placed a distance r apart. The normal to the current loops, $\hat{n}_1$ and $\hat{n}_2$ make angles θ_1 and θ_2 with the line joining the loops. The vectors $\hat{n}_1$ and $\hat{n}_2$ and the line joining the centers are co-planar as shown in figure. [3+3+1]

- Find the mutual inductance $\mathcal{M}$ of this system of current loops. Assume the radius of each loop is much smaller than the distance between loops.
- Find the force $\mathbf{F}$ between the two loops.
- How would the force be different if the currents were reversed in one or both of the current loops ?



Solution:

- Choose the z direction along $\hat{n}_1$. Magnetic moment of loop 1 is $IA\hat{z}$. Then the field due to loop 1 at loop 2 is

$$\mathbf{B} = \frac{\mu_0}{4\pi} \frac{IA}{r^3} (2 \cos \theta_1 \hat{\mathbf{r}} + \sin \theta_1 \hat{\theta})$$

Then the flux through the loop 2 is

$$\begin{aligned} \phi_{21} &= A\hat{n}_2 \cdot \mathbf{B} \\ &= \frac{\mu_0}{4\pi} \frac{A^2}{r^3} (2 \cos \theta_1 \hat{n}_2 \cdot \hat{\mathbf{r}} + \sin \theta_1 \hat{n}_2 \cdot \hat{\theta}) I \\ &= \frac{\mu_0}{4\pi} \frac{A^2}{r^3} (2 \cos \theta_1 \hat{n}_2 \cdot \hat{\mathbf{r}} + \sin \theta_1 \hat{n}_2 \cdot \hat{\theta}) I \\ &= \frac{\mu_0}{4\pi} \frac{A^2}{r^3} (2 \cos \theta_1 \cos \theta_2 - \sin \theta_1 \sin \theta_2) I \end{aligned}$$

Thus the mutual inductance is

$$\begin{aligned} \mathcal{M} &= \frac{\mu_0}{4\pi} \frac{A^2}{r^3} (2 \cos \theta_1 \cos \theta_2 - \sin \theta_1 \sin \theta_2) \\ &= \frac{\mu_0 A^2}{8\pi r^3} [3 \cos(\theta_1 + \theta_2) + \cos(\theta_1 - \theta_2)] \end{aligned}$$

- The force on the second loop due to the first one is given by

$$\begin{aligned} \mathbf{F} &= \nabla (\hat{\mathbf{m}}_2 \cdot \mathbf{B}) \\ &= \nabla \left(\frac{\mu_0}{4\pi} \frac{I^2 A^2}{r^3} (2 \cos \theta_1 \cos \theta_2 - \sin \theta_1 \sin \theta_2) \right) \\ &= \frac{\mu_0 I^2 A^2}{4\pi} \left(-\frac{3\hat{\mathbf{r}}}{r^4} (2 \cos \theta_1 \cos \theta_2 - \sin \theta_1 \sin \theta_2) \right. \\ &\quad \left. - \frac{\hat{\theta}}{r^4} (2 \sin \theta_1 \cos \theta_2 + \cos \theta_1 \sin \theta_2) \right) \end{aligned}$$

- The direction of $\mathbf{F}$ is reversed if (only) one of the currents is reversed.

Question 6: In free space, an electric field is given by

[2+2+2+1]

$$\mathbf{E} = \hat{x}E_0 [\cos(kz - \omega t) + \cos(kz + \omega t)].$$

- Find the magnetic field $\mathbf{B}$.
- Determine the poynting vector and its time average over one cycle.
- Find the time averaged electric and magnetic energy densities.
- Does this field represent a wave? If so, what kind of wave does it represent?

Solution:

- a) Electric field $\mathbf{E} = \hat{x}E_0 [\cos(kz - \omega t) + \cos(kz + \omega t)] = \hat{x}2E_0 \cos kz \cos \omega t$.

The magnetic field is given by

$$\begin{aligned} B &= \int dt (-\nabla \times E) \\ &= \hat{y} \int dt (2E_0 k \sin kz \cos \omega t) \\ &= 2E_0 \frac{k}{\omega} \sin kz \sin \omega t \hat{y} \\ \therefore H &= \frac{2}{c\mu_0} E_0 \sin kz \sin \omega t \hat{y} \end{aligned}$$

- b) The poynting vector

$$\mathbf{S} = \frac{1}{\mu_0 c} E_0^2 \sin 2kz \sin 2\omega t \hat{z}$$

and

$$\langle S \rangle = \frac{1}{T} \int_0^T \mathbf{S} dt = 0$$

- c) The electric and magnetic energy densities are

$$\langle u_E \rangle = \epsilon_0 E_0^2 \cos^2 kz$$

and

$$\langle u_M \rangle = \frac{E_0^2}{c^2 \mu_0} \sin^2 kz = \epsilon_0 E_0^2 \sin^2 kz$$

- d) Yes. Its a standing wave.